Precambrian Research 228 (2013) 164–176

Contents lists available at SciVerse ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Paleomagnetism of ca. 2.3 Ga mafic dyke swarms in the northeastern Southern Granulite Terrain, India: Constraints on the position and extent of Dharwar craton in the Paleoproterozoic

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ARTICLE INFO

Article history:
Received 29 February 2012
Received in revised form
12 December 2012
Accepted 3 January 2013
Available online 26 January 2013

Keywords:
Paleomagnetism
Geochronology
Dykes
Southern Granulite Terrain
Dharwar craton
Paleoproterozoic

ABSTRACT

The pole positions reported for the mafic dykes occurring in the northeastern Southern Granulite Terrain and easterly trending dykes of Dharwar craton are alike while their ages differ by ca. 700 Ma confounding tectonic reconstruction. To resolve the discordance and to constrain the position of the Indian continent in the Proterozoic, paleomagnetic and geochronological studies have been carried out on the mafic dyke swam in the northeastern Southern Granulite Terrain. Two distinct pole positions are obtained, one from the group of Tiruvannamalai dykes as 27.7°S, 231.5°E (dp, dm = 12°, 14°) with corresponding mean declination (Dm) and inclination (Im) as 125° and −73.8° (κ, α95 = 22, 7.6°) respectively, and second from the East Coast Dykes as 2.32°S, 188.2°E (dp, dm = 5°, 8°) with corresponding mean Dm and Im as 88.9° and −33.8° (κ, α95 = 48, 7.1°) respectively. The Remanent Efficiency of Magnetization values between 0.01 and 0.08 suggest that the dykes have been cooled through Curie temperature of magnetite which is the carrier of magnetization. The baked contact test confirms that the Characteristic Remanent Magnetization is the primary magnetization. The Sm–Nd mineral-whole-rock isochron, from one of the Tiruvannamalai dykes, yielded an age of 2318 ± 60 Ma (MSWD = 1.9). This age is the time of intrusion of the Tiruvannamalai dykes and is similar, within errors, to the age of dykes in the Dharwar craton. Thus, the contiguity of Dharwar craton with the Northern Block of Southern Granulite Terrain, as early as 2.3 Ga ago, is established. The pole position thus obtained corresponds to a high-latitudinal position of Dharwar craton during early Paleoproterozoic.

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1. Introduction

The early continental crust grew and stabilized as cratons by the end of the Archaean that later joined to form proto continents and several such continents joined to form supercontinents (Taylor and McLennan, 1985; Jacobsen, 1988; Martin et al., 2005). The Indian subcontinent has been included in major supercontinent reconstructions, such as Columbia, Rodinia, Gondwana and Pangaea (Rogers, 1996; Rogers and Santosh, 2002; Zhao et al., 2004; Meert, 2012). Meert et al. (2010) suggested that Aravalli, Bastar, Bundelkhand, Dharwar and Singhbhum cratons were welded together at 2.5–2.6 Ga ago to form proto-India. However, the tectonic relationship between these Archaean cratonic nuclei and surrounding granulite terrains that recorded various Proterozoic magmatic and metamorphic events is not well understood (Mezger and Cosca, 1999; Meissner et al., 2002; Collins and Pisarevsky, 2005; Santosh et al., 2003, 2005).

The Archaean Dharwar craton is surrounded by two granulite terrains, the Eastern Ghats Mobile Belt (EGMB) to the east, and the Southern Granulite Terrain (SGT) to the south. The metamorphic grade of rocks within the Dharwar craton increases from greenschist facies in the north to granulite facies in the south. The northern part of the SGT has numerous shear zones, such as, Moyar, Bhavani, and Mettur (Javadi Hills), which were considered to be the boundary between the Dharwar craton and SGT (Drury et al., 1984; Ray et al., 2003). Based on the airborne magnetic and gravity survey Mishra and Kumar (2005) identified two dominant bands of thrust zones, namely, the Moyar shear zone and the Palghat–Cauvery shear zone (PCSZ). According to Bhaskar Rao et al. (1996), the larger of these two zones, the PCSZ, is an E–W trending shear zone and marks the southern margin of Dharwar craton. The PCSZ has been

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http://dx.doi.org/10.1016/j.precamres.2013.01.004

correlated by several authors with the Betsimiserka shear zone of Madagascar and interpreted as the Ediacaran–Cambrian suture (Collins et al., 2007, 2008; Raharimahefa and Kusky, 2009) associated to the final amalgamation of the Gondwana Supercontinent.

Before any reliable continental reconstructions can be addressed, one of the fundamental questions which still need to be answered is the paleo-position of the SGT during Paleoproterozoic. So far the position of the eastern Dharwar craton for the time period is well constrained, based on precise U–Pb geochchronology and paleomagnetic studies on ca. 2.4 Ga old dolerite dykes (Halls et al., 2007). In contrast, even though they show similar paleomagmetic trends, the dykes in the SGT were believed to be ca. 1.6 Ga, based on K–Ar dating of feldspars, (Venkatesh et al., 1987; Radhakrishna and Joseph, 1996; Halls et al., 2007) and therefore cannot be associated to the same dynamic event. To solve this discordance and to understand better the precise relationship between the low-grade Dharwar craton with the bordering high-grade terrain, dating the dykes within the SGT, south of the Dharwar craton, along with determination of paleopole position is crucial.

Furthermore, the ongoing efforts to reconstruct the land distribution prior to the Columbia supercontinent, i.e. earlier than ca. 1.8 Ga, are confounded by a lack of reliable paleomagnetic and geochronological data (Raub et al., 2007). Recently, Kumar et al. (2012) have proposed that the Dharwar craton lay very close (<10°) to the Slave province during the Paleoproterozoic. However, they could not conclusively relate the Dharwar craton to the Neoarchaean supercratons, Sclavia or Superia, due to lack of reliable paleomagnetic data. It is therefore important to have more datasets pertaining to Paleoproterozoic and older periods, in order to come up with a meaningful paleo-continental reconstruction. We report results of integrated paleomagnetic and geochronological studies on the mafic dykes of the northeastern SGT and its implication for defining the paleo-position and extent of the Dharwar craton.

2. Geological setting

Southern peninsular India mainly consists of the Archaean Dharwar craton, the Cuddapah basin of Proterozoic-age, the Eastern Ghats Mobile Belt and the high-grade Southern Granulite Terrain (Fig. 1a; Drury et al., 1984). The SGT is considered to be a mosaic of micro continents or crustal blocks that are juxtaposed along the shear zones (Chetty et al., 2003). Two major shear zones, E–W trending the Palghat–Cauvery in the north and NW–SE trending Achankovil in the south, divide the SGT into three separate units, (a) the Northern Block, (b) the Madurai Block, and (c) the Trivandrum Block (Fig. 1a; Drury et al., 1984). The 'Northern Block' mainly consists of orthopyroxene-bearing granulites and hornblende-biotite gneiss. Syenite and carbonatite intrusive bodies (ca. 750 Ma) occur along the Mettur shear zone (MSZ) (Fig. 1b) (Morolov et al., 1975; Kumar et al., 1998; Miyazaki et al., 2000; Santosh et al., 2005). The Madurai Block, lying south of the Palghat–Cauvery shear zone and north of the Achankovil shear zone, is characterized by hypersthene ± saphirine ± corundum-bearing granulites, pelitic and mafic granulites, hornblende-biotite gneiss, calc-silicates and intrusive rocks of ca. 550 Ma age (Janardhan, 1999; Tsunogae and Santosh, 2003). Further south of the Achankovil shear zone, khondalite, charnokite, metapelites and metacarbonates with strong imprints of Pan-African metamorphism constitute the Trivandrum Block (Fonarev et al., 2000; Santosh et al., 2003, 2009).

The Madras–Tiruvannamalai (MT) terrain, a triangular landmass on the map of the 'Northern Block', is bounded by Mettur and Palghat–Cauvery shear zones to the northwest and south, respectively and the Bay of Bengal to the east. The foliation planes observed in the MT terrain are predominantly trending in the NE–SW direction and are parallel to the trend of the Mettur shear zone. This terrain consists of hypersthene-bearing granulites (also known as charnockites), hornblende-biotite gneiss, granites and mafic dyke swarms.

The hypersthene-bearing granulites were formed around temperatures close to 800 °C and pressures of 7–9 kbars (Weaver, 1980; Condie et al., 1982; Raith et al., 1982; Manglik, 2006) and experienced isothermal decompression (Mohan and Jayananda, 1999; John et al., 2005). The granitoid gneisses of the Dharwar craton are considered as protolith of these granulites (Srivastava and Kanishkan, 1975–1976).

Based on Sm–Nd and Rb–Sr whole-rock isotope studies on the hypersthene-bearing granulites of the transition zone, the age of regional metamorphism is suggested to be 2500 Ma (Vidal et al., 1988). The massive charnockite from the north of Cauvery shear zone has yielded a Sm–Nd whole-rock isochron age of 2490 ± 60 Ma and U–Pb zircon ages ranging from 2454 to 2556 Ma (Bhutani et al., 2007a,b). The early Paleoproterozoic granites are recognized around Gingee, Tiruvannamalai and Tirukovilur along with younger migmatitic granite within the MT terrain and the Rb–Sr age of migmatitic granites were reported to be 2254 ± 60 Ma (Balasubrahmanyan et al., 1978–1979).

2.1. Mafic dyke swarms of Southern India

The dykes within the peninsular gneiss are found to be widely distributed and cross-cutting dyke swarms are exposed around Hyderabad, SW of Cuddapah basin, around Bangalore and in NW Dharwar craton. However, such occurrences of dyke swarms are sparse within the SGT (Geological Survey of India, 1981; Fig. 2 in French and Heaman, 2010). Dykes of southern India are classified under three categories; (a) metamorphosed but undeformed meta-dolerite and meta-norite dykes; (b) swarms of un-metamorphosed dolerite dykes; and (c) alkaline dykes (Ikramuddin, 1968,1974; Ikramuddin and Stueber, 1976). Based on the precise U–Pb ages, mafic dykes are grouped into three age ranges of 2369–2365 Ma, 2221–2209 Ma and 2181–2177 Ma (French and Heaman, 2010), while the alkaline dykes were found to be younger (1192 Ma and 1027 Ma) (Pradhan et al., 2008, 2010). The mafic dyking within the eastern Dharwar craton spans nearly 200 Ma (2369–2177 Ma), recording multiple period of mafic magmatism and is similar to the dyking pattern documented for the Paleoproterozoic Scourie dykes in Scotland (Heaman and Tarney, 1989; French and Heaman, 2010). All the dykes occurring within the Dharwar craton and the SGT were broadly classified as picrites to basaltic andesite and tholeiitic in compositions with characteristic negative Nb anomaly (Murty et al., 1987; Radhakrishna et al., 1991; Rao et al., 1990; Radhakrishna and Joseph, 1998; Halls et al., 2007; Radhakrishna, 2009; French and Heaman, 2010). The dykes occurring within the high grade SGT are more enriched in silica compared to the dykes of Dharwar craton (Radhakrishna and Joseph, 1998; French and Heaman, 2010). Petrographically, dykes in SGT are characterized by the clouding of plagioclase, which is considered to be related to CO_2 metasomatism during carbonatite and syenite intrusions along the shear zones (Radhakrishna and Joseph, 1996; Halls et al., 2007). Paleomagnetic and geochronological studies on mafic dyke swarms that occur on either side of the Mettur (Javadi Hills) shear zone could help test whether this shear zone is a lithospheric-scale suture zone between Dharwar craton and high-grade granulites or it is just a crustal fault.

Several paleomagnetic studies on the dykes of the peninsular gneisses resulted in three different pole positions: (i) 10°N, 211.4°E from the Anantpur alkaline dyke of 1027 Ma age (Pradhan et al., 2010); (ii) 15.7°N, 56.9°E from the easterly trending Bangalore dolerite dykes (2365 Ma) (Halls et al., 2007); and (iii) 29.7°S, 261°E from the N–S trending Harohalli alkaline dykes (1192 Ma)

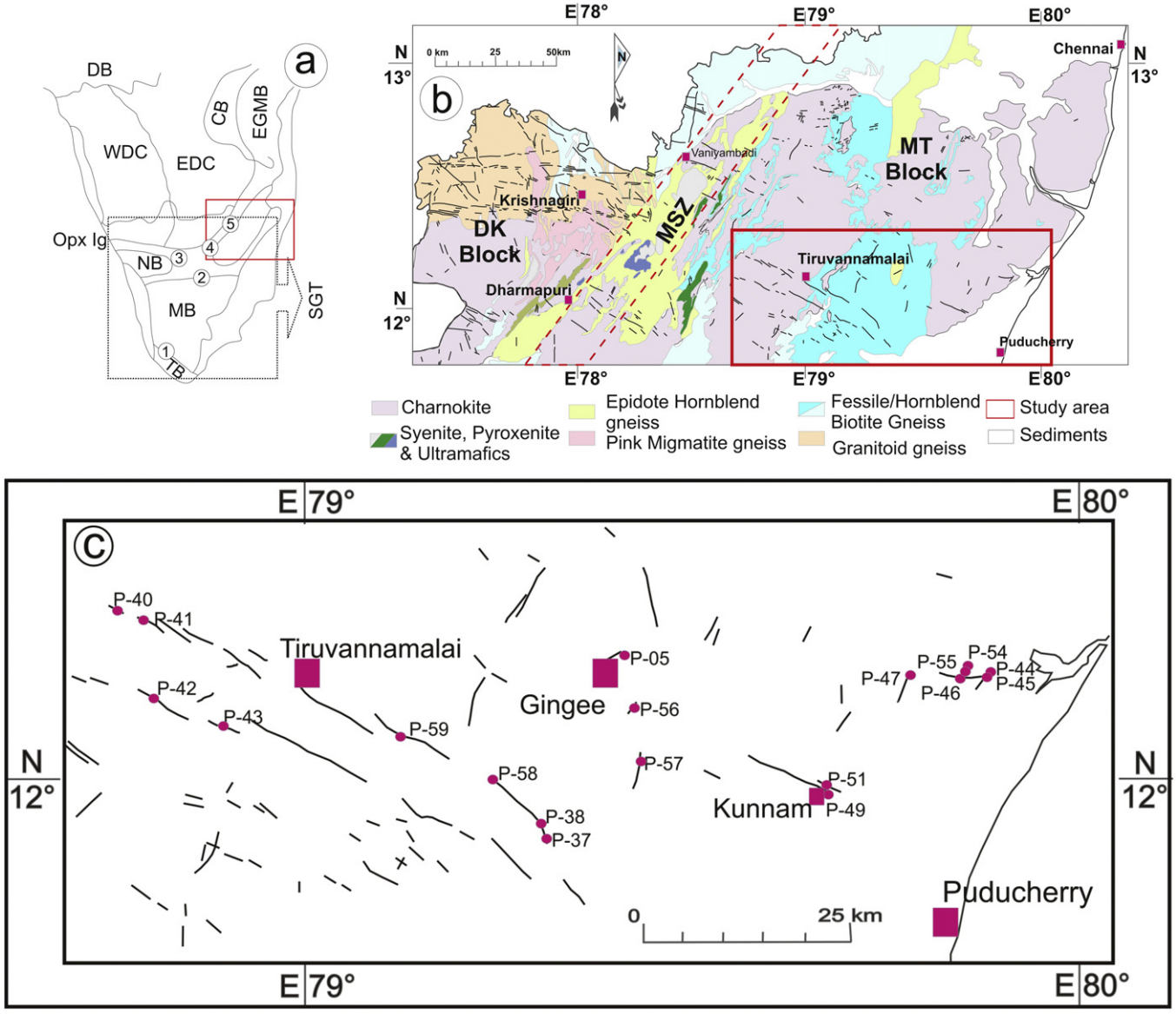


Fig. 1. (a) Simplified tectonic map of South India showing the study area (modified after Drury et al., 1984). DB, Deccan basalt; CB, Cuddapah basin; EGMB, Eastern Ghat mobile belt; EDC, Eastern Dharwar craton; WDC, Western Dharwar craton; Opx Ig, Orthopyroxene isograd; SGT, Southern granulite terrain; NB, Nilgiri block; MB, Madurai block; TB, Trivandrum block; 1, Achankovil; 2, Palghat–Cauvery; 3, Moyar–Bhavani; 4, Salem–Attur; 5, Mettur shear zone, (b) The distribution of major rock types in the study area; DK block, Dharmapuri–Krishnagiri block; MSZ, Mettur Shear Zone; MT block, Madras–Tiruvannamalai block, and (c) The details of the sampling sites in Madras–Tiruvannamalai block.

(Pradhan et al., 2008). Four additional pole positions have been identified from the high grade terrain of southern India but they have poor age constraints (Radhakrishna and Joseph, 1996) and therefore need to be taken cautiously. Their positions and ages, are: (i) the Agali-Anaikatti swarm (75°N, 153°E; ca. 2000 Ma); (ii) the Dharmapuri swarm (79°N, 101°W; 1800–1750 Ma); (iii) the northern Kerala dykes (51°N, 130°E; 1700–1650 Ma); and (iv) the Tiruvannamalai swarm (18.8°S, 125.2°W; 1650 Ma). However, Halls et al. (2007) included some of the sites from Tiruvannamalai dyke swarm in the compilation of easterly trending dykes of the Dharwar craton.

The dykes within the MT terrain of SGT are wide, massive, and trend predominantly NW–SE with subordinate NE–SW orientations and are compositionally uniform, believed to represent a single magmatic episode along conjugate sets of fractures (Radhakrishna and Joseph, 1998; Subramanian and Selvan, 2001). An integrated geochronological and paleomagnetic study of these dykes is important to provide a precise timing of intrusion of dykes, with additional paleomagnetic data required to constrain the paleo-position of India during Proterozoic.

3. Field work, sampling and petrography

Representative samples of mafic dykes occurring in the MT terrain around Tiruvannamalai, Gingee and near the southeastern coast of Chennai were collected for the present study (Fig. 1c). The width of the dykes varies between 20 m and 60 m and they preserve sharp chilled margins with coarsening of the grain size towards the centre, at places. Fresh, oriented samples of dykes were collected using a portable drill corer from deep rock-quarries except for a few that were collected from in situ surface exposures. Eleven dykes were sampled at 15 sites (Fig. 1c) and in some cases the same dyke was sampled at more than one site. All together 108 in situ core samples were collected and 99 representative core specimens

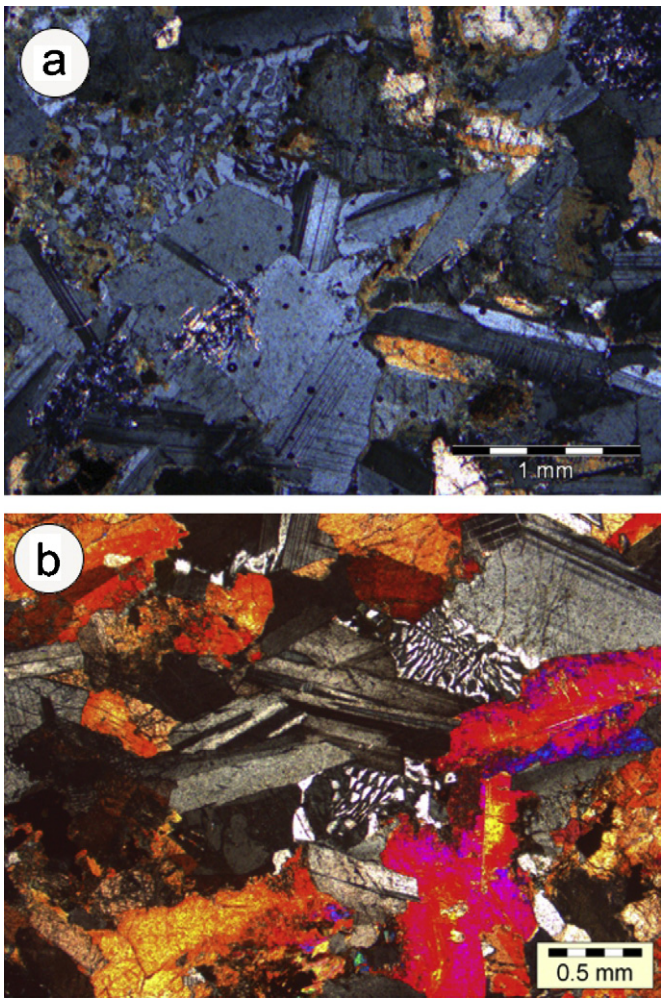


Fig. 2. Photomicrographs of mafic dykes from the MT block depict, (a) intergrowth of quartz and feldspar resulting in granophyric texture (top left corner) and interlocking laths of plagioclase and pyroxene and (b) laths of plagioclase are partially enclosed in pyroxene grains and granophyric texture is also seen. These features indicate that the dykes preserve igneous texture and minerals.

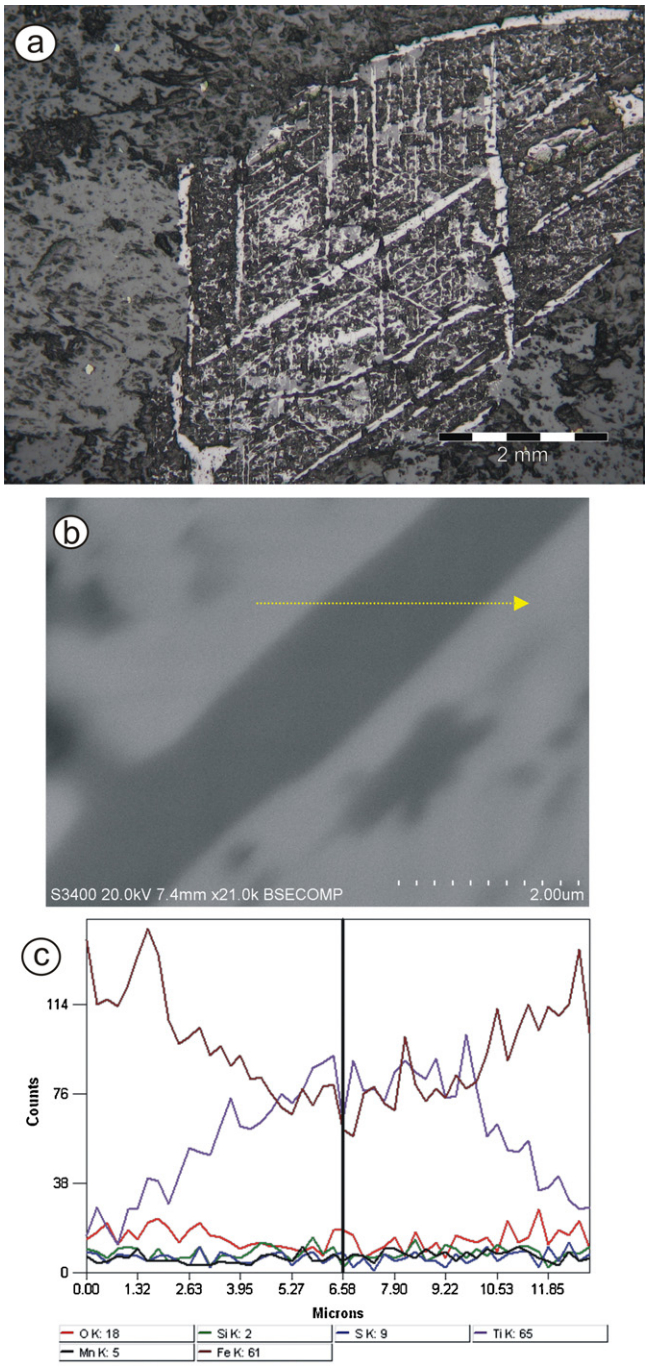


Fig. 3. (a) Exsolution lamellae of ilmenite along octahedral planes of magnetite under reflected light and (b) Secondary Electron (SE) image of polished section where ~5 μm-thick band of titano-magnetite exsolution is clearly resolved. (c) Energy Dispersive X-rays (EDX) line scan for elements O, Si, S, Ti, Mn and Fe along the yellow line shown in (b).

were analyzed after shaping and polishing the core to the required dimensions (2.5 cm diameter and 2.2 cm length).

The dykes are medium to coarse grained, mainly consist of plagioclase, pyroxene and opaque minerals, and preserve original igneous textures. The plagioclase and pyroxene commonly show ophitic or sub-ophitic texture (Fig. 2a and b) with distributed subhedral or anhedral grains of opaque minerals. Brownish clouding of plagioclase is commonly seen in all the dyke samples. Magnetite is the dominant opaque mineral while sulfides are present in trace quantities. In dykes from the Tiruvannamalai area, magnetite shows perfect exsolution lamellae of titano-magnetite along the octahedral cleavage planes that could have been developed during slow cooling (Fig. 3a–c). Exsolution lamellae are not observed in magnetite from the dykes occurring close to the east coast.

4. Geochronology

Based on the petrographic studies, samples were selected for mineral separation and isotope studies. Each sample was ground to 125–177 μm size. The heavy minerals were separated using bromoform after magnetite was removed with a hand magnet. Magnetic separator (Frantz^{TM} model LB-1) was used at currents starting from 0.1 A to 1.8 A to separate paramagnetic minerals with various susceptibilities. Pure pyroxenes and plagioclase feldspars devoid of inclusions and alterations were handpicked, under a stereomicroscope, from magnetic fractions at 0.3–0.5 A and non-magnetic fraction at 1.8 A, respectively. The mineral samples were digested following conventional digestion techniques using the double distilled suprapure HF, HCl and HNO_3 acids. About one-third of precisely weighed digested samples were spiked with tracer solutions enriched in ^{87}Rb, ^{84}Sr, ^{152}Sm and ^{150}Nd isotopes for determination of concentrations of Rb, Sr, Sm and Nd by the isotope dilution method. The Rb and Sr were separated in 2 N HCl using Bio-Rad AG50 W X8, 200–400 mesh, cation exchange resin,

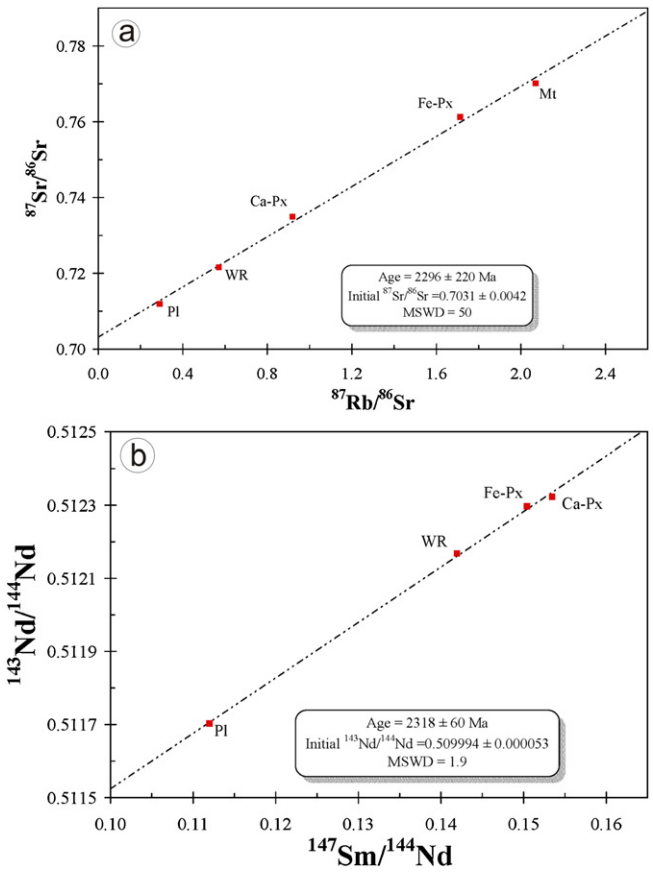


Fig. 4. The Rb–Sr and Sm–Nd mineral-whole-rock isochron for a NW-trending dyke from Tiruvannamalai region.

while the Nd and Sm were separated using the HDEHP coated Teflon resin at 0.3 N and 0.4 N HCl, respectively. Detailed procedure for sample digestion and chemical separation of Rb, Sr, Sm and Nd for isotope studies is given in Anand and Balakrishnan (2010). The Rb, Sr, Sm and Nd isotope ratios were measured using a Thermal Ionization Mass Spectrometer (Thermo-Finnigan, model-Triton) at the Department of Earth Sciences, Pondicherry University. The isotopic compositions of Sr and Nd were corrected for mass fractionation using the internal normalizing ratios of $^{86}Sr/^{88}Sr = 0.1194$ and $^{146}Nd/^{144}Nd = 0.7219$. To correct the isotope ratios of the spiked fraction for mass-fractionation, correction factors (per amu) determined by multiple measurements of isotopic standards for Rb (−0.0033), Sr (0.00040), Sm (0.00247) and Nd (0.00039) were applied. The mean $^{87}Sr/^{86}Sr$ ratio determined on SRM987 during the course of the analyses is 0.710243 ± 5 (1σ) ($n = 40$) and $^{143}Nd/^{144}Nd$ ratio measured on AMES is 0.511970 ± 4 (1σ) ($n = 25$) (reference values: SRM 987 $^{87}Sr/^{86}Sr = 0.710244$ and AMES $^{143}Nd/^{144}Nd = 0.511968 \pm 4$ (Govindaraju, 1994)).

The results of Rb–Sr and Sm–Nd isotopic studies on different mineral separates and whole-rock are given in Table 1. A regression line in the Rb–Sr isochron plot of two different pyroxenes, plagioclase, magnetite and whole-rock of a dyke from Tiruvannamalai (P-58) indicates large scatter and yield an errorchron with a slope equivalent to an age of 2296 ± 220 Ma (MSWD = 50; Fig. 4a). In contrast, the Sm–Nd isochron plot for the two pyroxenes, plagioclase, and whole-rock yield an isochron age of 2318 ± 60 Ma (MSWD = 1.9; Fig. 4b). The higher scatter in the Rb–Sr system is probably caused by the post-emplacement alteration or low-temperature metamorphism. However, Rb–Sr age is within the uncertainties of the more precise Sm–Nd age, thereby confirming that the Sm–Nd system is inherently a more robust system, less

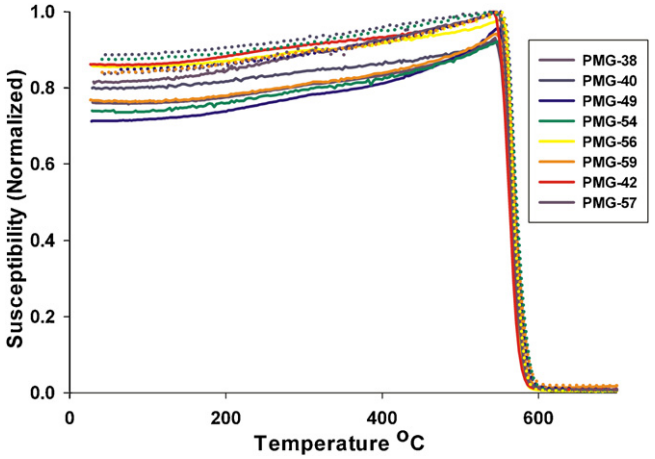


Fig. 5. Thermo-magnetic χ–T curves showing magnetite as dominant remanence carrier (solid and dotted lines represent heating and cooling curves respectively).

affected by post-emplacement alteration. The minerals that define the Sm–Nd isochron are of igneous origin as confirmed by petrographic study. Hence, this Sm–Nd isochron age represents the time of crystallization of these minerals from the magma and could be regarded as the age of dyke intrusion.

5. Rock magnetism and Paleomagnetism

The magnetic susceptibility and Natural Remanent Magnetization (NRM) were measured with the help of MS2B dual frequency sensor (Bartington Instruments, England) and Molspin spinner magnetometer (England) at the Department of Physics, Pondicherry University. For one set of samples, the susceptibility was measured using AGICO MFK1-FA, multi-function Kappa-bridge (fields of 200 A/m at 976 Hz) at Indian Institute of Geomagnetism, Navi Mumbai. The demagnetization of the specimens was carried out at the Centre for Earth Sciences Studies, Trivandrum and Indian Institute of Geomagnetism, Navi Mumbai, for selected samples. During treatment all samples were stored in a field-free space to avoid contaminations. Temperature dependent low-field (300 A/m, 875 Hz) magnetic susceptibility was measured up to 700 °C on KLY-4 Kappa-bridge in an Ar atmosphere to avoid oxidation of the magnetic minerals.

5.1. Rock magnetic results

The measured magnetic properties of the dykes are given in Table 2. The NRM values of the dykes vary between 0.5 and 1.8 A/m, except P-49, which shows higher (~3.8 A/m) NRM values. The parameters of low frequency susceptibility (χ_{lf}) and Saturation Isothermal Remanent Magnetization (SIRM) are related to magnetic mineral concentration; χ_{lf} varies in a narrow range from 91 to 138×10^{-7} m^3 kg^{-1}, indicating the uniform distribution of magnetic minerals in all dyke samples. The Q-ratio varies between 0.19 and 3.23 for the Earth's magnetic field of 39.79 A/m. The value of χ_{fd}% varies in a very narrow range with very low values (~2%; Table 2), indicating the absence of super-paramagnetic grains. From the coercivity values (Table 2), the Soft-IRM (Isothermal Remanent magnetization) is observed to be dominant component over the Hard-IRM, suggesting occurrence of low coercive magnetite as the dominant ferrimagnetic mineral.

From the temperature dependent low-field magnetic susceptibility it is shown that both heating and cooling χ–T-curves (susceptibility versus temperature) are reversible (Fig. 5). The susceptibility values remain constant up to a temperature of 580 °C

Table 1
Isotopic data for the separated minerals from dyke P-58 near Tiruvannamalai in Southern Granulite Terrain.

Minerals/samples	Rb_{conc} (ppm)	Sr_{conc} (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$	$\times 10^{-6}$ (2σ)	Sm_{conc} (ppm)	Nd_{conc} (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$	$\times 10^{-6}$ (2σ)
Magnetite	21.87	30.76	2.0696	0.770150	±14	–	–	–	–	–
Fe-Pyroxene	27.82	47.23	1.7131	0.761259	±8	4.09	14.39	0.1504	0.512297	±16
Ca-Pyroxene	22.45	70.89	0.9186	0.734971	±7	3.29	11.36	0.1534	0.512323	±15
Plagioclase	26.66	265.71	0.2904	0.711928	±11	3.72	17.60	0.1120	0.511703	±7
Whole rock	32.10	162.98	0.5707	0.721610	±7	3.29	12.28	0.1419	0.512168	±6

Note: The 2σ error of 1.0% and 0.5% as have been assumed for $^{87}Rb/^{86}Sr$ and $^{147}Sm/^{144}Nd$ respectively during calculation of age.

Table 2
Rock magnetic results of the selected dyke-samples from the northeastern part of SGT.

Sample	χ_{lf} (10^{-7} m^3 kg^{-1})	χ_{fd}%	ARM (10^{-5} A m^2 kg^{-1})	NRM (A/m)	SIRM (A/m)	REM	SIRM (10^{-5} A m^2 kg^{-1})	Soft IRM 10^{-5} A m^2 kg^{-1}	Hard IRM 10^{-5} A m^2 kg^{-1}	$S_{-300\,mT}$
P-49	91.0	2.04	144.8	3.765	52.4	0.072	10,223	4707	26	0.997
P-51	98.2	2.18	49.3	0.910	44.9	0.020	8048	4384	229	0.972
P-57	137.7	2.50	72.2	0.614	89.2	0.007	15,651	11,711	598	0.962
P-38	111.2	2.28	51.1	1.176	72.0	0.016	13,212	176	451	0.966
P-58	110.4	1.99	54.7	0.685	80.3	0.009	13,339	9851	94	0.993
P-59	85.6	2.16	41.0	0.876	68.2	0.013	10,357	7141	69	0.993
P-46	87.2	2.28	71.7	1.361	56.4	0.024	9472	4264	336	0.964
P-54	108.8	2.08	58.1	1.305	62.8	0.021	9781	4692	161	0.984
P-56	183.4	1.97	73.5	1.783	130.0	0.014	15,131	14,580	273	0.982
P-40	114.9	2.22	40.2	0.698	40.8	0.017	7658	4872	84	0.989
P-42	102.6	2.27	35.2	0.461	54.5	0.008	8494	5590	229	0.973

Where χ_{lf}: low frequency susceptibility, $\chi_{fd}\% = (\chi_{lf} - \chi_{hf})/\chi_{lf}$, ARM: anhysteretic remanent magnetization, NRM: Natural Remanence Magnetization, SIRM: Saturation Isothermal Remanent Magnetization, REM: Ratio of NRM and SIRM, IRM: Isothermal Remanent Magnetization, $S_{-300\,mT} = IRM_{-300\,mT}/SIRM$.

and sharply decrease to almost zero with increasing temperature, indicating magnetite as the dominant magnetization carrier. The $S_{-300\,mT}$ (defined as $IRM_{-300\,mT}/SIRM$), very close to unity (ranges from 0.96 to 1.0) indicates the presence of ferrimagnetic magnetite and titano-magnetite grains in the samples (Basavaiah and Aniruddha, 2004). As most of REM (Remanent Efficiency of Magnetization = NRM/SIRM) values range between 0.01 and 0.08, it is inferred that the dykes have acquired magnetization during continuous cooling through the Curie temperature and are not affected by any subsequent low temperature metamorphic activity. Though the REM values for P-42, P-57 and P-58 dykes are < 0.01, they are very close to 0.01 (Table 2). Since none of the REM values exceeds 0.9, it is suggested that the dyke samples were not struck by lightning (Wasilewski and Warner, 1988).

5.2. Alternating Field (AF) and thermal demagnetization

A set of 24 specimens taken from 11 dykes was selected for a pilot study. Alternating Field (AF) demagnetization was carried out in small incremental steps (2.5–100 mT) in order to characterize the magnetic behaviour and to choose the best demagnetization procedure. A total of 16 specimens were subjected to 15 and 19 incremental steps of thermal demagnetization (100–700 °C). Commonly, goethite (hard magnetic domain) transforms to fine grain hematite in range of 250–370 °C, (Gehring and Heller, 1989; Dekkers, 1990). The samples that were hard to demagnetize using AF were heated to 150 °C to demagnetize the hard magnetic domains (e.g. goethite; Curie temperature 110–130 °C; Dunlop, 1990). Then they were subjected to 15 steps of incremental AF demagnetization (2.5–100 mT) to prevent the transformation to another hard magnetic domain.

During AF demagnetization most of the samples demagnetize up to ~60% at around 10–20 mT (Fig. 6a). A few specimens from the sites, P-41, P-42, P-49, P-58 and P-59, show very hard magnetization against increasing AF up to 100 mT. At low fields up to 40 mT these samples show no demagnetization behaviour, demagnetizing only at high fields between 50 and 100 mT (Fig. 6b).

From the normalized NRM versus temperature plot (Fig. 7), it is observed that there is hardly any change in the NRM value for the dyke samples P-49, P-51, P-58, up to 450 °C while it decreases gradually between 450 and 550 °C and reduces to zero level at around 575 °C (a characteristic of discrete unblocking); this demagnetising behaviour is characteristic of magnetite. However, for the other representative specimens, P-40, P-41, P-42 and P-54 (Fig. 7), the

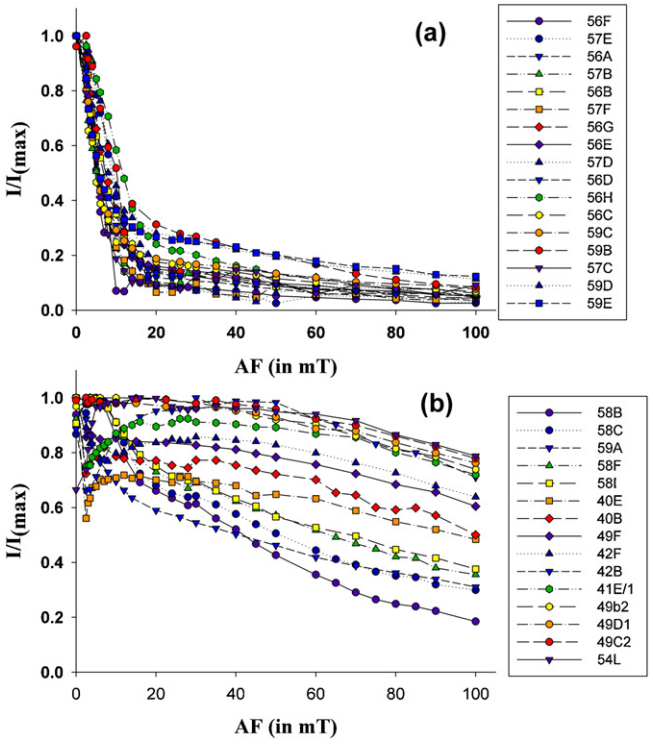


Fig. 6. The demagnetization response of NRM with increasing Alternating Field. The group of specimens indicates the presence of, (a) soft magnetic minerals and (b) hard magnetic minerals.

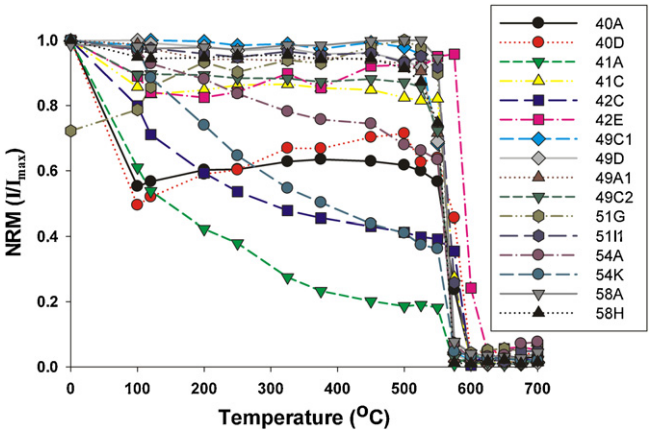


Fig. 7. The demagnetization response of NRM with increasing temperature.

normalized NRM decreases gradually between 100 and 550 °C and specimens are completely demagnetized at 575 °C (a characteristic of distributed unblocking).

5.3. Baked contact test

The Baked contact test (BCT) proposed by Brunhes (1906) and modified by Everitt and Clegg (1962) mainly serves to compare the direction of remanent magnetization in an igneous rock unit with that of the host rocks. The in situ drill core samples were collected for BCT from the country rock at three different sites. Variation in the primary magnetization directions are given in Table 3 and Fig. 8.

The width of the dyke at site P-58 is ~20 m and is intruded into a hornblende-biotitie gneiss (quartz, orthoclase, plagioclase, biotitie, hornblende ± apatite ± epidote ± sphene). The dyke can be traced over ~10 km in NW–SE direction. The country rock at a distance of 1.5 m from the contact yields similar magnetization directions, while the country rock sampled at a distance of 2.5 m, 35 m and ~10 km shows different magnetization directions (Fig. 8a) indicating that the episode of dyke intrusion was not contemporaneous with the cooling of the crust to below the Curie temperature (T_c).

The mafic dyke at site P-42 is ~15 m wide, at the outcrop and intrudes into migmatized quartzo-feldspathic gneiss. The samples of country rock at a distance of 15 m and 100 m from the contact show similar magnetization directions which seem to have been affected by the dyke intrusion, while the country rock at a distance of 2 km shows different magnetization directions (Fig. 8b).

Dyke at site P-49 belongs to East Coast Dyke (ECD) group and is intruded into quartzo-feldspathic gneiss. The magnetization directions at distances 1 m, 3 m, 5 m, 12 m, and 20 m are consistent and show different magnetization compared to the dyke, whereas, the country rock at a distance of 2 km yield a different magnetization. Apparently this dyke does not indicate a baked contact and intruded into country rock probably below the T_c, but nevertheless has not been affected by a post-emplacement regional metamorphism.

5.4. Virtual Geomagnetic Poles

Characteristic Remanent Magnetization (ChRM) directions are determined using the principal component analysis (PCA). All directions are based on linear segments defined by 6–10 steps and a mean angle deviation (MAD) value < 4. The Zijderveld diagrams and stereographic projections for representative dyke samples are given in Figs. 9 and 10, respectively. The mean site directions (Dm and Im) and Virtual Geomagnetic Poles (VGPs) were calculated individually and the site ChRM values are summarized in Table 4. Based on this analysis the samples define two groups with distinct ChRM and these are referred to as Tiruvannamalai Dykes (TD) and East Coast Dykes (ECD) (Fig. 11). One site (P-56; NE–SW trending dyke) shows a direction that is roughly antipodal to the main TD group and is comparable with the ChRM and VGPs of the N-S trending Harohalli alkaline dykes (Pradhan et al., 2008). The ChRM directions of dykes from the Tiruvannamalai area reported by Radhakrishna and Joseph (1996) and Venkatesh et al. (1987) are similar to the directions obtained during the present study. The mean ChRM directions (Dm and Im) obtained from the individual sites for the TD group of dykes during the present study are also comparable with the mean ChRM directions reported by Halls et al. (2007). The mean paleomagnetic directions, based on a compilation of all the existing results along with present data, for the dykes of Tiruvannamalai region, are found to be Dm = 125° and Im = −73.8° (α95 = 7.6°; κ = 22). The corresponding position of paleomagnetic pole at 27.7°S; 231.5°E (dp/dm = 12°/14°) (Table 4) places India at paleo-latitude of approximately 60°S.

The ChRM determined for the ECD group of dykes (Fig. 1C) is comparable with that of the dykes from Kunnam area reported by Venkatesh et al., 1987 ("*Group a*") and Radhakrishna and Joseph, 1996 (site T12, T13). The combined paleomagnetic vectors are found to be Dm = 88.9° and Im = −33.8° (α95 = 7.1°; κ = 48) with the VGP at 2.32°S; 188.2°E (dp = 5°; dm = 8°) (Table 4) places India at paleolatitude of 18.6°S. The clear distinction in the ChRMs of the dykes based on the primary magnetic vector (Dm and Im) is shown in Fig. 11.

6. Discussion

In the attempt to understand better the paleo-position of the MT terrain our results show that two distinct paleomagnetic directions

Table 3
Results (Declination, D and Inclination, I) of the baked contact test performed at three different sites in the MT block.

Distance:-		Dyke	1.5 m	2.5 m	35 m	10 km
P-58	D	174.9	1.48	228.8	114.5	179.4
	I	-75.9	-69.87	-15.34	-13.22	74.36

Distance:-		Dyke	15 m	100 m	2 km
P-42	D	130.2	74.08	44.07	169.5
	I	-72.1	-80.58	-68.43	-46.43

Distance:-		Dyke	1 m	3 m	5 m	12 m	20 m	2 km	5 km
P-49	D	93.1	282.16	326.7	86.8	280.7	302.7	203.9	209.79
	I	-48.5	-76.62	-81.12	-80.37	-76.39	-79.03	-22.31	-26.27

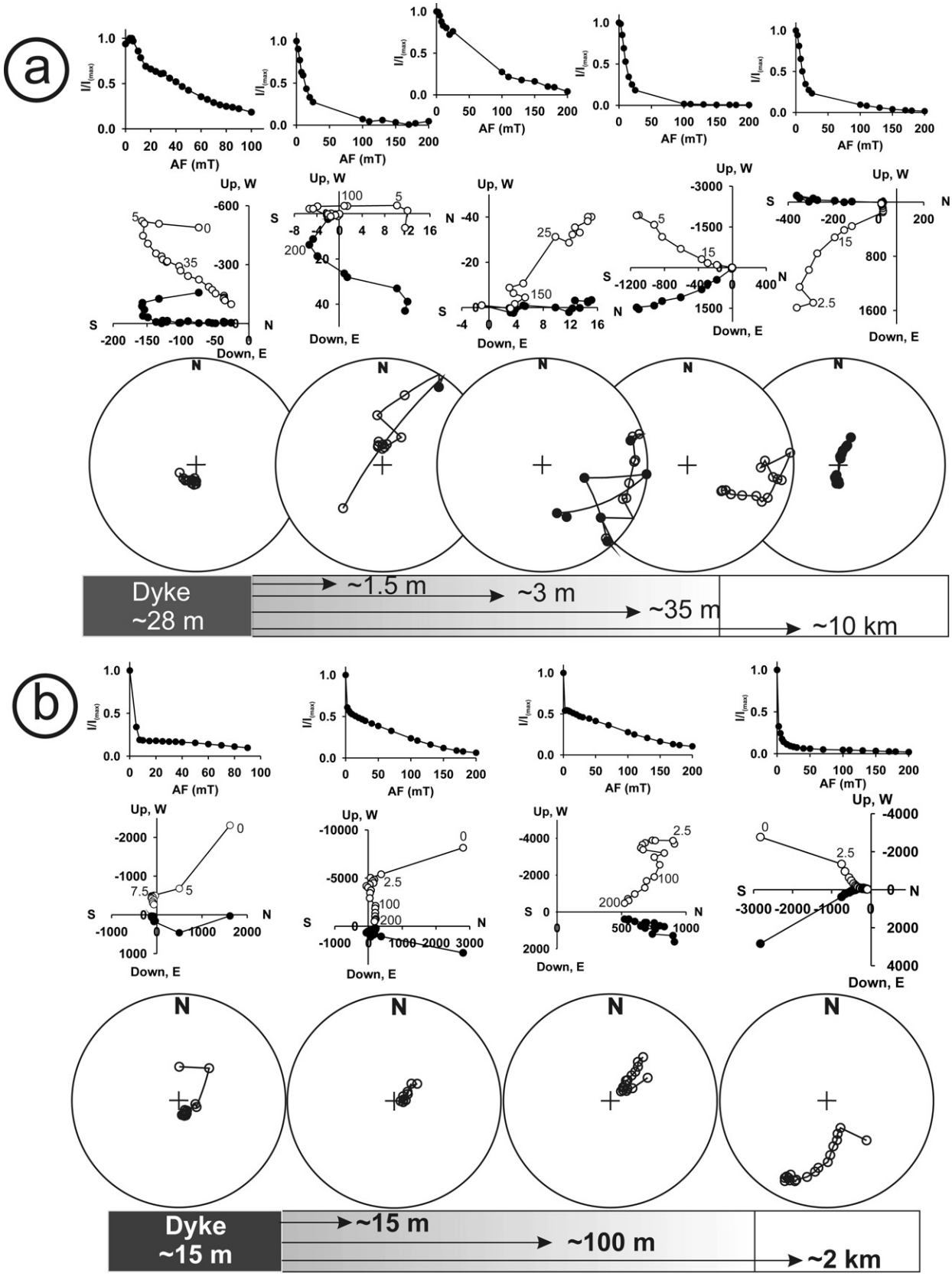


Fig. 8. Results of the baked contact test performed near the site P-58 (a) and P-42 (b) of Tiruvannamalai Dykes (Table 3). Top row in (a) and (b) shows the AF demagnetization depicting the decay of NRM; Zijderveld diagrams are plotted in the middle row and corresponding equal-area stereo-plots are given in the bottom row. Open circles in the Zijderveld diagrams represent vertical components and solid circles represent horizontal components; while in the stereoplots, the open circles are for the up (negative) and solid circles are for the down (positive) inclinations.

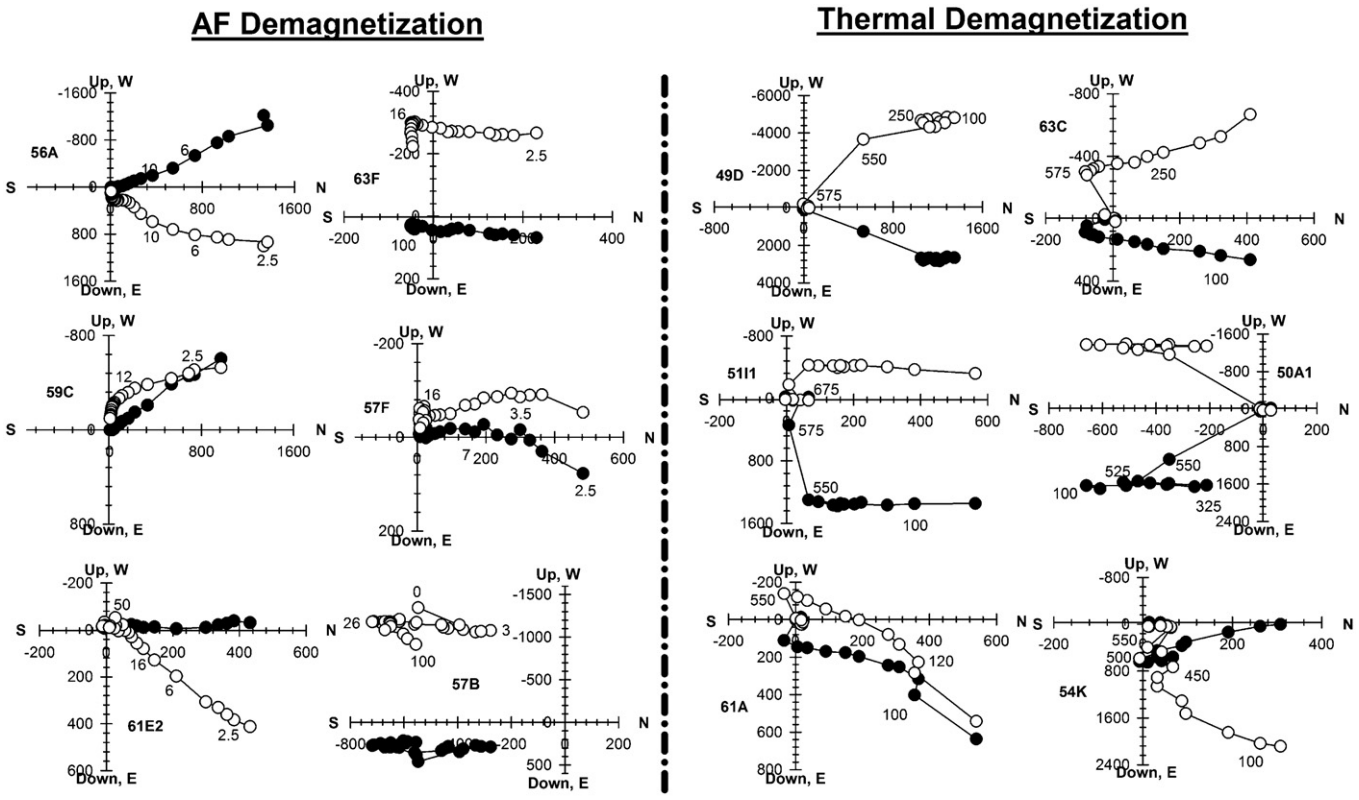


Fig. 9. Zijderveld diagrams showing the migration of vectors, (a) during AF demagnetization and (b) during thermal demagnetization. Open circles are vertical components and solid circles are horizontal components.

are present in the Tiruvannamalai dykes (TD) group and the East Coast Dykes (ECD) group. To make sure that these are consistent and comparable to other dykes in the Dharwar craton a Sm–Nd mineral isochron age of TD group has also been determined.

6.1. Tiruvannamalai dykes (TD)

The time of regional granulite facies metamorphism in the northeastern SGT is inferred to be shortly after the emplacement of the igneous protoliths at around 2500 Ma, (Vidal et al., 1988; Raith et al., 1990; Friend and Nutman, 1992; Nutman et al., 1992; Harris et al., 1994; Janardhan et al., 1994, 1996; Srikantappa, 1996; Bhutani et al., 2007a,b). We interpret our newly obtained Sm–Nd mineral isochron of 2318 ± 60 Ma for the Tiruvannamalai dyke as the time of igneous crystallization of the dykes in the MT block and that the previously reported K–Ar whole-rock age of 1630 ± 30 Ma (Radhakrishna and Joseph, 1996) for the dykes from this region is incorrect due to partial or complete resetting of the K–Ar isotope system by low temperature events. Ghosh et al. (2004) and Bhutani et al. (2007b) have also inferred widespread magmatism north and south of the Palghat Cauvery shear zone during 2500–2300 Ma based on U–Pb zircon ages. However, due to superimposed

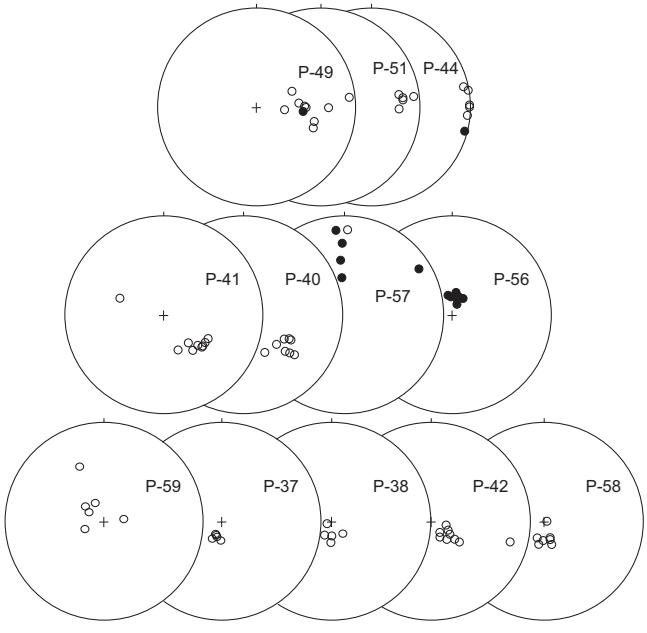


Fig. 10. Stereographic projections of the ChRM for individual sites from MT terrain. Open circles indicate upward and solid circles indicate downward inclinations.

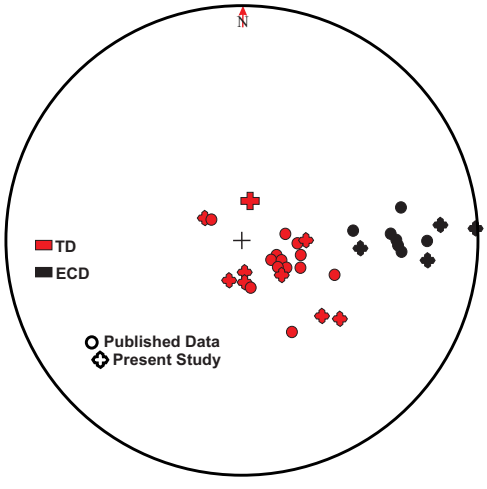


Fig. 11. The magnetic vectors (Table 4) for the dykes occurring around east coast and Tiruvannamalai region of the MT terrain are grouped as ECD and TD, respectively.

Table 4
Paleomagnetic data for individual sites of Madras–Tiruvannamalai (MT) terrain in Southern Granulite Terrain (SGT).

Site	Trend	*N*	*n*	Lat (°N)	Long (°E)	Dm (°)	Im (°)	κ	α95 (°)	λ (°)	*Φ* (°)	dp (λ)	dm (Φ)
Group TD													
P-37[*]	**150**	**5**	**5**	**12.06**	**79.35**	**197.9**	**−76.1**	**716**	**2.9**	**36.9S**	**269.2 E**	**5**	**5**
P-38[*]	**140**	**6**	**4**	**12.05**	**79.33**	**174.5**	**−79.3**	**88**	**9.8**	**32.6S**	**257 E**	**18**	**19**
P-58[*]	**115**	**6**	**7**	**12.11**	**79.25**	**174.9**	**−75.9**	**100**	**6.1**	**38.7S**	**256.3 E**	**10**	**11**
P-59[*]	**115**	**5**	**4**	**12.16**	**79.16**	**302.2**	**−74.8**	**100**	**9.2**	**3.7S**	**283 E**	**15**	**17**
P-56	45	8	8	12.16	79.36	11.3	75.5	291	3.3	38.9S	266 E	6	6
P-57	30	6	5	12.1	79.45	357.3	26.8	8	28.2	86.7N	28 E	17	31
P-40[*]	**130**	**12**	**8**	**12.26**	**78.88**	**128.4**	**−46.2**	**70**	**6.7**	**39.5S**	**194.6 E**	**6**	**9**
P-41[*]	**135**	**11**	**8**	**12.26**	**78.9**	**133.2**	**−51.8**	**88**	**5.9**	**38.8S**	**200.9 E**	**6**	**8**
P-42[*]	**125**	**9**	**8**	**12.2**	**78.9**	**130.2**	**−72.1**	**129**	**4.9**	**31.3S**	**229.9 E**	**8**	**9**
P-43[*]	**120**	**5**	**2**	**12.16**	**78.95**	**89.1**	**−68.1**	**91**	**26.5**	**8.9S**	**219.6 E**	**37**	**45**
T2 [@,*]		**6**	**1**	**12.2**	**79.08**	**79.7**	**−74.7**	**–**	**–**	**5.8S**	**230.7**	**–**	**–**
T3 [@,*]		**7**	**–**	**12.09**	**78.92**	**129.7**	**−73.5**	**481**	**2.8**	**9.9S**	**232**	**–**	**–**
T4 [@,*]		**7**	**4**	**12.06**	**79.01**	**114.4**	**−67.9**	**92**	**9.6**	**24.6S**	**219.8**	**–**	**–**
T5 [@,*]		**7**	**6**	**12.05**	**79.03**	**306.5**	**−76.7**	**275**	**4.1**	**3.4N**	**279.2**	**–**	**–**
T7 [@,*]		**7**	**–**	**12.05**	**79.08**	**92.1**	**−70.7**	**69**	**7.3**	**11S**	**233.3**	**–**	**–**
T8 [@,*]		**6**	**–**	**12.11**	**79.1**	**121**	**−78.1**	**109**	**11.9**	**29.9S**	**232**	**–**	**–**
T9 [@,]		7	–	12.22	78.8	96.6	−0.8	111	6.4	6.5S	167.8	–	–
T10 [@,*]		**6**	**2**	**12.22**	**78.8**	**120.7**	**−72.1**	**57**	**33.7**	**26.7S**	**227.3**	**–**	**–**
T-11 [@,*]		**7**	**3**	**12.27**	**78.91**	**109.6**	**−55.8**	**45**	**18.5**	**22.9S**	**203.4**	**–**	**–**
Gr-c VIII [#,*]		**2**	**–**	**12.17**	**78.93**	**151.0**	**−54.0**	**130**	**10.0**	**–**	**–**	**–**	**–**
Gr-c IX [#,*]		**3**	**–**	**12.2**	**78.87**	**102.0**	**−69.0**	**76**	**9.0**	**–**	**–**	**–**	**–**
Group mean (15 dykes)				12.2	78.5	**125**	**−73.8**	22	7.6	**27.7S**	**231.5E**	12	14
Group ECD													
P-44	35	6	6	12.23	79.88	86.7	−1.3	120	7.0	3.13N	171 E	4	7
P-46[*]	**90**	**5**	**3**	**12.2**	**79.83**	**95.7**	**−22.5**	**419**	**6.0**	**7.95S**	**180 E**	**3**	**6**
P-49[*]	**135**	**13**	**8**	**12.06**	**79.83**	**93.1**	**−48.5**	**30**	**10.3**	**8.53S**	**198.3 E**	**9**	**14**
P-51[*]	**140**	**7**	**5**	**12.06**	**79.66**	**85.1**	**−17.1**	**121**	**7.0**	**2.9N**	**179 E**	**4**	**7**
P-54	95	4	2	12.23	79.83	91.9	34.4	190	18.2	2.2N	151 E	12	21
Gr-a I [#,*]		**6**	–	–	–	**78.0**	**−32.0**	**40**	**9.0**	–	–	–	–
Gr-a II [#,*]		**6**	–	–	–	**89.0**	**−35.0**	**60**	**7.0**	–	–	–	–
Gr-a III [#,*]		**6**	–	–	–	**87.0**	**−38.0**	**98**	**6.0**	–	–	–	–
Gr-a IV [#,*]		**5**	–	–	–	**94.0**	**−33.0**	**100**	**6.0**	–	–	–	–
Gr-a V [#,*]		**3**	–	–	–	**87.0**	**−37.0**	**230**	**5.0**	–	–	–	–
T-12 [@,*]		**7**	–	**12.08**	**79.66**	**90.1**	**−23.1**	**550**	**2.8**	**2.6S**	**181.4 E**	–	–
T-13 [@,*]		**6**	–	**12.06**	**79.66**	**84.3**	**−51.1**	**74**	**7.9**	**1.6S**	**201.9 E**	–	–
Group mean (5 dykes)				12.1	79.7	**88.9**	**−33.8**	48	7.1	**2.32S**	**188.2 E**	5	8

N: number of oriented drilled cores, *n*: number of specimens analyzed for mean primary directions, Lat and Long: site latitude and longitude, Dm and Im: mean declination and mean inclination, κ: Fisher's precision parameter, α95: ellipse of 95% confidence, λ and *Φ*: VGP latitude and VGP longitude respectively. dp (λ), dm (Φ): correction in semi minor and semi major axis. Values in bold type are included in calculating the group mean.

[*] The sites considered for pole estimation.
[#] Data from Venkatesh et al. (1987)
[@] Data from Radhakrishna and Joseph (1996).

metamorphic events, so far no primary remanent magnetization has been reported from the south of Palghat Cauvery Shear zone.

Interestingly, the age obtained from a dyke of the MT block (2318 ± 60 Ma) is within the analytical uncertainties of the U–Pb baddeleyite ages (2365–2368 Ma) for the Bangalore dyke swarm (Halls et al., 2007; French and Heaman, 2010) and it provides an opportunity to compare the paleomagnetic directions of these two regions. Halls et al. (2007) reported paleo-pole of 15.7°N and 56.9°E for the ca. 2.37 Ga old Bangalore dyke swarm which is statistically comparable to the pole position 27.7° S; 231.5° E obtained here for the TD group dykes of the MT block. In addition, the ChRM and the pole position reported for the Dharmapuri dykes (Radhakrishna and Joseph, 1996) are similar to the secondary component-B observed for the dykes occurring near Bangalore and Harohalli (Halls et al., 2007; Pradhan et al., 2008).

These pole-positions are also comparable with the Harohalli dolerite dykes (Dawson and Hargraves, 1994) which yielded an indistinguishable emplacement age of 2370 ± 230 Ma.

The results from our study are similar to the pole positions reported by Venkatesh et al. (1987), who studied the dykes from Tiruvannamalai and concluded that the pole position was at 31°N and 54°E and the paleo-latitude of the Indian sub-continent was at 66°N. Radhakrishna and Joseph (1996) inferred the pole position for dykes from the same vicinity as 18.8°S and 125.2°W. Both these estimates are similar to the present study; however, age of these dykes has now been revised by us to ca. 2.32 Ga from ca. 1.6 Ga based on the Sm–Nd isochron.

6.2. East Coast Dykes (ECD)

The paleo-pole position obtained from ECD is 2.3°S and 188.2°E. This is distinct from the nearby TD group of dykes. To interpret this dichotomy two explanations are possible, either the dykes were emplaced during a magmatic event of different age or the ECD were separated significantly in space from the TD region at the time of intrusion. However, to rationalize the geographic separation a complex explanation has to be found, since there is no major tectonic zone separating the TD and ECD regions. Petrographic studies from the Tiruvannamalai dykes (TD) show exsolution lamellae of ilmenite within titano-magnetite (Fig. 3), whereas the ECD lacks such features, indicating a somewhat different crystallization history and we interpret that these two dyke swarms not be cogenetic.

The ChRM and VGP for the ECD group are in good agreement with the pole position (9.7°S; 182°E) of Oddanchatram anorthosite

(Satyanarayana et al., 2003) and Anantapur alkaline dykes (10°N and 211°E; 1027 ± 13: U–Pb age) (Pradhan et al., 2010). The primary pole position of the Oddanchatram anorthosite was compared with that of Wajrakarur kimberlites (Rb–Sr age 1090 ± 20 Ma, Kumar et al., 1993) and Tirupati dykes (Table 3 in Satyanarayana et al., 2003) whose ages are not constrained. However, Ghosh et al. (2004) reported a U–Pb Zircon age of 563 ± 9 Ma for the Oddanchatram anorthosite which corresponds to growth of zircons during the Pan-African regional igneous and metamorphic events (Söderlund et al., 2002). Therefore, similar pole positions for igneous rocks from different geographical regions of the Indian subcontinent do not necessarily means that they are of the same age. On other hand, pole positions of the similar age rocks are also reported to be varying significantly. Although there is similarity in ChRM and VGP of ECD with the Anantapur alkaline dykes, they differ in their chemical composition. Therefore, in the absence of precise age of the dykes found near the east-coast (ECD) of the MT terrain their inferred paleo-positions would remain tentative.

6.3. Paleo-tectonic reconstruction of the Northern Block of SGT

The Southern Granulite Terrain (SGT), exposing a vast tract of high-grade rocks ranging from Ultra High Temperature to Ultra High Pressure granulites, has been a key terrain in various attempts of paleo-supercontinent reconstruction from early Paleoproterozoic Columbia to Neoproterozoic Gondwana (Crawford, 1974; Hoffman, 1991; Fitzsimons, 2000; Meert, 2002, 2012; Cenki and Kriegsman, 2005). Though the superimposition of various deformation and metamorphic events makes it difficult to correlate this terrain geologically with other coeval and contiguous terrains of erstwhile supercontinents, the polymetamorphic character of the terrain is unambiguously caused by several episodes of break-up and assembly. Though the SGT extends up to the orthopyroxene isograd to the north, first identified by Fermor (1936) and thereby known as the Fermor line, it is only the Neoproterozoic Palghat Cauvery Shear Zone that has been correlated with Bongolava-Ranotsara shear zone or Betsimisaraka shear zone in Madagaskar in the west and with the boundary between Rayner and Napier complexes of Antartica in the east (Harris et al., 1994; De Wit et al., 1995; Meissner et al., 2002; Reeves et al., 2002; Cenki and Kriegsman 2005). Recently, it has also been proposed to be a suture and the site of Mozambique ocean closure culminating the Gondwana amalgamation during the Neoproterozoic (Janardhan, 1999; Rambeloson et al., 2003; Cenki and Kriegsman, 2005; Collins et al., 2007; Raharimahefa and Kusky, 2009; Santosh et al., 2009,2012; Plavsa et al., 2012). In this context, the high grade terrain north of Palghat–Cauvery Shear zone and the south of the Fermor line has not been well understood. Drury et al. (1984) had visualized this 'Northern Block' as part of the integral granulite terrain that is dissected by crustal scale shear zones. However, another school of thought maintained that the 'Northern Block' is a deeper part of the Archaean Dharwar Craton, based on the structural continuity of the Dharwar craton (Gopalakrishna et al., 1986; Naha and Srinivasan, 1996; Meissner et al., 2002). The present study is first to report primary magnetization preserved in ca. 2.3 Ga old dykes from the 'Northern Block' that yields a comparable paleopole position with the contemporaneous dykes from the Dharwar craton. Though a similar paleopople position was reported earlier from here (Radhakrishna and Joseph, 1996), due to inherent limitations of the chronometers used to date these dykes, the correct age could not be estimated resulting in incorrect correlations. These present results also resolve the outstanding issues related to the tectonic significance of the shear zones within the 'Northern Block', particularly the Mettur shear zone which lies to the west of the MT terrain, the region of present study.

Mettur shear zone in the 'Northern Block' is a major linear structure which is marked by alkaline-carbonatite rock association. It has been variedly interpreted and its tectonic significance is under debate. Harinarayana et al. (2006), based on a magnetotelluric survey, suggested that the Dharwar craton and the MT terrain were juxtaposed along the less resistive Mettur shear zone. Seismic survey (Reddy et al., 2003) suggests differences in crustal structure with a deep seated fault along the Mettur shear zone. Based on the airborne magnetic and gravity survey (Mishra and Kumar, 2005) it has been proposed that there is a N–S compression between the Dharwar craton and SGT, probably resulting in the Mettur shear zone. During the present study of the TD group of dykes, lying east of the Mettur shear zone, we obtained a pole-position of 27.7°S and 231.5°E similar to that obtained for the easterly trending dykes of the Dharwar craton (Halls et al., 2007; French and Heaman, 2010). The TD dykes dated as ca. 2318 ± 60 Ma are coeval with the Dharwar dykes, therefore, establishing the contiguity of the Dharwar craton with the Northern Block of SGT as early as 2.3 Ga ago. Based on the above findings, it is suggested that Mettur shear zone, within the Northern Block of SGT may represent a lithospheric scale fault or shear zone and is not a suture zone between disparate terrains.

7. Conclusion

Based on the rock magnetic results (i.e. REM values) and a baked contact test, it is shown that the ChRM present in the TD dykes of SGT is primary magnetization and was acquired during the cooling of the dykes through Curie temperature, mainly of magnetite, and there is no evidence of later thermal events affecting the dykes of this region.

These dykes occurring in the northeastern part of Southern Granulite Terrain (SGT) are grouped as TD and ECD based on paleomagnetic results. The Sm–Nd mineral isochron age of 2318 ± 60 Ma represents the time of igneous intrusion and cooling of TD. The pole position and age is similar with the pole position and U–Pb baddeleyite ages reported earlier (Halls et al., 2007; French and Heaman, 2010) for the easterly trending dykes of the Dharwar craton. Therefore it is established that the 'Northern Block' of SGT was contiguous with the Dharwar craton at 2.3 Ga ago and consequently the high-grade rocks in this region could represent the deeper section of the Dharwar craton. The available palaeomagnetic data along with the present studies yield paleo-pole position which places India at high latitudes (∼60°) during early Paleoproterozoic.

Acknowledgements

The authors acknowledge funding by the Department of Science and Technology, India. The Central Instrumentation Facility, Pondicherry University, is acknowledged for extending access to the SEM facility. Dr. T. Radhakrishna is thanked for providing access to the paleomagnetic laboratory facilities during the initial stages of this study. We are grateful to Prof. J.G. Meert and an anonymous reviewer for their thorough and thoughtful review of the earlier version of this manuscript. The manuscript is hugely improved due to the in-depth review by Prof. Wouter Bleeker and by the editorial suggestions and review by Dr. Randall Parrish. We thank Prof. R. Ramesh and Dr. Robert Buchwaldt for helping us improve the language of the manuscript.

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